§7 companion note: a de Finetti sweep behind the n_{eff} panel, and a pinned false-fire scenario

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Date	2026-09-08
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This note answers two requests on §7: (1) give a de Finetti *sweep range* behind the $n_{\text{eff}} \approx 1.6$ panel rather than a single point, and (2) pin down the naive-plug-in false-fire scenario, correcting an earlier "∼90%" figure. All numbers are from `iclr/sims/section7_sim.py` at commit `38ab67d` (seed `20260907`, purely additive read-outs; the martingale construction and seed are unchanged from commit `5c225ad`).

1 The de Finetti n_{eff} sweep ($N = 6$)

Setup. The two-atom de Finetti model in Panel (a) induces within-item co-failure $\rho = \text{sep}^2$ across the swept grid $\text{sep} = \texttt{linspace}(0, 0.9, 10)$. Under the equicorrelation structure this model realizes, the Kish effective sample size for $N = 6$ judges is

$$n_{\text{eff}}(\rho) \;=\; \frac{6}{1 + (6-1)\,\rho}.$$

ρ	n_{eff}
0.0000	6.0000
0.0100	5.7143
0.0400	5.0000
0.0900	4.1379
0.1600	3.3333
0.2500	2.6667
0.3600	2.1429
0.4900	1.7391
0.6400	1.4286
0.8100	1.1881

Table 1: $n_{\text{eff}}(\rho)$ across the swept grid $\rho = \text{sep}^2$, $\text{sep} \in \texttt{linspace}(0, 0.9, 10)$.

Range. As ρ sweeps $[0, 0.81]$,

$$n_{\text{eff}} \in [1.19,\ 6.00],$$

with the minimum 1.19 at $\rho = 0.81$ and the maximum 6 at $\rho = 0$ (independent judges).

Key point. The empirical FailureScope anchor $n_{\text{eff}} = 1.6350$ ($\bar{\varphi} = 0.5340$, $N = 1253$ items, 6 judges) sits at $\rho = \bar{\varphi} = 0.5340$ on this very curve, and the consistency check

$$n_{\text{eff}}(0.5340) = \frac{6}{1 + 5 \cdot 0.5340} = 1.6350$$

holds exactly. So the ≈ 1.6 figure is *not* a free parameter — it is the image of the observed mean pairwise co-failure $\bar{\varphi}$ under the same de Finetti curve the power panel already sweeps. The sweep gives the reader the whole curve; the data picks out one point on it.

2 The naive-plug-in false-fire scenario, pinned

Honest correction. The earlier "$\sim$90%" figure came from a pre-fix version — the non-martingale `np.roll` adjacent-pairing construction. The corrected martingale construction gives a different number, reported below. I am flagging this discrepancy rather than smoothing over it.

Canonical scenario, pinned to explicit parameters. $K = 6$ judges; `n_items` $= 300$; `n_streams` $= 2000$; betting fraction $\lambda = 0.2$; `eps` $= 0.02$; drift = base failure rate sweeping linearly $0.25 \to 0.60$ across the stream; given each item's rate the judges fail independently (Bernoulli), so there is *zero* conditional co-failure — only shared marginal drift within an item. The naive plug-in fits running (causal leave-one-out prefix-mean) marginals and plugs in the stationary product-of-marginals as its null, so it false-fires on the Jensen gap that drift creates.

Canonical result. Naive-plug-in false-fire ≈ 0.743 (0.74); the margins-free cross-item process holds size at 0.0000.

Regime	Drift	Naive false-fire	Margins-free size
gentle	$0.30 \to 0.50$	0.166	0.000
current	$0.25 \to 0.60$	0.72–0.74 (canon. 0.743)	0.000
steep	$0.15 \to 0.75$	0.9995	0.000

Table 2: Sensitivity to drift steepness. K, `n_items`, `n_streams`, λ all fixed; only the drift endpoints change. The "current" naive figure is a Monte-Carlo band with canonical value 0.743.

Reading. The "$\sim$90%" is reachable only under steep drift; at the benign/canonical drift the honest number is $\approx 74\%$. The load-bearing claim is invariant across the whole range: *the naive plug-in false-fires catastrophically under benign marginal drift with zero true co-failure, while the margins-free cross-item process holds size at* 0.000*.* Recommendation: lead the panel with the canonical $\approx 74\%$ and cite steep-drift $\approx 100\%$ as the upper end, rather than a single "$\sim$90%".

3 One caveat on the martingale gate

The gate is clean at short horizons: under a $\delta = 0$ true null,

$$\mathbb{E}[e_T] = 1.0003\ (N = 2), \quad 1.0006\ (N = 3), \quad 0.9987\ (N = 4).$$

At $N = 400$, $\mathbb{E}[e_T] = 0.9638$ with median 0.1299 — the heavy right-skew the construction anticipates (needs more replications for the mean to settle), not a soundness failure. Worth noting since it is the softest number in the gate.

Closing: what I am least sure about

Whether to headline the dramatic steep-drift $\approx 100\%$ or keep the honest canonical $\approx 74\%$. I lean toward canonical $\approx 74\%$ with the sensitivity table shown, so the panel cannot be accused of scenario-tuning. Ask for your read.